

RE-AL THEMATIC SERIES

RESEARCH ARTICLE

Defining large-scale arid island vegetation recovery targets through evaluating a reference ecosystem within an archipelago extent

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Island ecological restoration presents challenges in establishing historical frames and reference ecosystems. Our study takes place in the Galapagos Islands and focuses on North Seymour (NS), recognized as a reference ecosystem for Baltra Island, which has suffered significant degradation. We assessed NS's arid plant community, comparing it with early twentieth-century records. A first survey identified that dominant woody species on NS include native *Bursera graveolens*, *Cordia lutea*, *Castela galapageia*, *Parkinsonia aculeata*, and *Opuntia echios*, while *Cleome viscosa* is the only exotic species registered. A second survey divided both islands into evenly distributed plots, recording adults and juveniles of the five dominant species from the first survey. NS showed a structured community with higher species richness and density compared to Baltra. *Castela galapageia* prevailed in disturbed areas on Baltra. Juvenile regeneration was scarce on both islands. NS exhibited high species richness throughout, while Baltra had only two areas with similar high richness. Co-occurrence analysis revealed significant associations between *P. aculeata* and *B. graveolens* on both islands. However, Baltra's network displayed missing links, indicating the need for restoration efforts. In conclusion, the structured plant community of NS can serve as one of the reference ecosystems needed for framing a restoration model for Baltra.

Key words: Baltra, *Bursera graveolens*, co-occurrence network, Galapagos, North Seymour, *Parkinsonia aculeata*, restoration targets

Implications for Practice

- Close islands can be used as reference ecosystems for ecological restoration within an archipelago.
- Smaller-scale sampling of the vegetation community can inform how to later extensively sample the dominant species.
- Large-scale surveys effectively represent the dominant plant species structure.
- Baltra Island, in the Galapagos, has undergone serious deterioration of its plant communities.

Introduction

It is of general knowledge that ecological restoration requires, as a first asset, information pertaining to the original landscape's key attributes, which is commonly inaccessible. Thus, defining reference ecosystems is a valuable approach for framing a reference model to restore areas that lack the original, unknown components (Aronson et al. 1993; Gann et al. 2019; FAO et al. 2023). These reference models should encompass a wide range of ecosystem attributes, including the absence of threats, species composition, community structure, physical conditions, ecosystem function, and external exchanges (Gann et al. 2019).

In plant-dominated systems, species composition, their density, and their structure serve as the foundation for establishing restoration objectives (Stanturf et al. 2014; Durbecq et al. 2020). In optimal scenarios, data regarding the original plant system can be accessed through herbaria records or literature (Anderson 2005). However, it is quite common that the available information is insufficient and only intermediate states of the ecosystem can be recovered (Holl 2020).

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Island restoration poses particular challenges that hinder efforts to recover their areas. The use of reference ecosystems is complex, as few other places behave and maintain similar ecological conditions (Lincoln et al. 2018). Thus, reference ecosystems are usually obtained through historical frames of the same site, which frequently can be insufficient and poorly documented (White & Walker 1997; Clewell et al. 2004; Lincoln et al. 2018). For example, due to the long-standing degradation on Hainan Island, the historical frame includes primarily non-native species, which are now included in restoration targets (Wang et al. 2021).

The Galapagos Verde 2050 (GV2050) is a program of the Charles Darwin Foundation (CDF) focused on the ecological restoration of the Galapagos Archipelago (Jaramillo et al. 2015, 2020). Through a stage-scale restoration procedure (Bakker et al. 2018), the program tests different technologies. This aims to address how to increase dozens of native plants' survival and growth in multiple study sites of the archipelago to scale-up interventions later (Negoita et al. 2021). Since 2013, the GV2050 has been an independent project aimed at restoring the arid vegetation of Baltra Island by recovering its plant populations and their structure (Tapia et al. 2019).

Baltra is by far the most degraded island in the Archipelago, due to decades of anthropogenic disturbances (Gibbs 2013, 2016). Close to the island is North Seymour, a smaller island that has suffered less degradation, which might work as a potential reference system. The restoration objectives for Baltra were defined in a first action plan for Baltra and the South Plaza Islands by scientific advisors of the CDF and Galapagos National Park Directorate (GNPD) (Jaramillo et al. 2017a, 2017b). The main focus of these objectives was to increase the presence of woody and cacti species, as they are the most physiognomic in the landscape and the main resources (i.e. food and shelter) for local wildlife, such as land iguanas and birds (Fig. S1). Also, focus is on taller vegetation, as currently Baltra herbaceous strata are dominated by the exotic *Cleoma viscosa* (Cleomaceae) (CDF 2023).

The knowledge gathered through the GV2050 program will be used to create guidelines for the GNPD to restore the island to its full extent. This increase in plant diversity is a necessary follow-up action by the GNPD (DPNG 2014). As the governmental agency responsible for the island restoration, they have previously eliminated goats and cats (Balseca 2002; Phillips et al. 2005), reintroduced iguanas (Cayot & Menoscal 1994), and will now complement these efforts with the eradication of rats. The GNPD has established that conservation efforts in the Galapagos should focus on maintaining ecosystem resilience and structure rather than replicating historical conditions, favoring ecological restoration over rehabilitation unless impractical, and avoiding recreation projects (DPNG 2014). In that tune, plant restoration activities on Baltra should start by replicating the community structure and composition of a reference system such as North Seymour. Furthermore, the restoration activities should strategically cover areas that can still accommodate tourist demand, while also considering current and future infrastructure needs (Tecno-Ambiente 2021).

Over the past decade, based on expert opinion, the GV2050 program has tested several combinations of species and restoration tools in small-scale areas close to the Baltra airport or the route to Santa Cruz (Fig. 2; Tapia et al. 2019; Negoita et al. 2021). The tools correspond to water-saving technologies such as hydrogel and Groasis Waterboxx, among others. Now, aiming to scale-up these efforts, questions about Baltra Island's pre-degradation condition need to be addressed in order to establish improved restoration objectives. The GV2050 is preparing a second version of the restoration plan for the period 2024–2029. This plan includes the objective of creating a reference model (Fig. S2) using various reference ecosystems with different degree of degradation and several complementary studies that will guide restoration efforts in Baltra. More specifically, one endeavor is to describe key ecosystem attributes, such as the species composition and the structure of the island's plant community.

Understanding the past vegetation of Baltra presents two primary challenges: limited historical data and biological differences between islands in the Galapagos. Early naturalists describing Galapagos flora were witnesses to the scarcity of plants in dry areas (Lloyd 1964; Charles Nelson & Porter 2011), probably mediated by centuries of foraging by introduced goats. Accordingly, little is known about the original plant composition and structure of Baltra's landscape. For example, the local Charles Darwin Research Station Herbarium has over 30,000 plant vouchers, but only 20 records from before 1990 are available for Baltra island (Mauchamp & Aldáz 1997; Jaramillo et al. 2017a, 2017b; CDF 2023). The limited literature available suggests that due to the low rain on the island, most vegetation has always been grasses, drought-tolerant bushes, thorny scrubs, small deciduous trees and cacti (Table S1) (Stewart 1911; Alpert 1946). Additionally, given the ecological and fragmented nature of the Galapagos land masses, it is thought that many lineages have experienced adaptive radiation, concluding in several endemic species (Porter 1984; Fernández-Mazuecos et al. 2020; Cerca et al. 2023). On the archipelago scale, of the 542 indigenous plant species, 252 are endemic (Jaramillo et al. 2017a, 2017b). All these features translate into the Galapagos Islands having a unique set of biological elements not typically shared between them, making finding adequate reference ecosystems challenging.

This paper aimed to set part of a scientific base for the development of a reference model and to enhance recovery targets for the restoration of Baltra, using the nearby island of North Seymour as one of the reference ecosystems. For this, key ecological questions regarding the plant composition, density, and dominance expected on Baltra were evaluated in random plots in a first survey on North Seymour. Then, after defining the most dominant species, in a second survey, North Seymour and Baltra islands were extensively sampled to evaluate the overall density and distribution of the five most dominant plant species, while also including their height structure. Both islands were also analyzed for the co-occurrence of dominant species, which can guide decisions on which species to plant together.

Methods

Study Sites and Background

The Galapagos Islands, situated along the equatorial line approximately 1000 km from mainland Ecuador, comprise 15 major islands and over a hundred small islands and islets (Snell et al. 1996). The unique climate of the archipelago is characterized by two distinct seasons, a hot season from January to May, marked by variable precipitation, and a cool season from June to December, characterized by scarce precipitation in the lowlands and consistent precipitation in the highlands (Trueman & D'Ozouville 2010). The El Niño Southern Oscillation (ENSO) has a large influence on the climate of the Galapagos. During El Niño years, there is a notable increase in precipitation and temperatures, whereas La Niña years bring about contrasting conditions (Tye & Aldáz 1999). Most plant species in Galapagos are dispersed by reptiles and birds (Forup et al. 2008; Heleno et al. 2011; Traveset et al. 2015). Then, it could be expected that, due to seed dispersion increasing with island proximity, near landmasses and islets would have similar plant communities.

Baltra Island

Baltra (0°27'07"S 90°16'36"W) is mainly a flat low-elevation island (<68 m a.s.l.) of 2700 ha, north of Santa Cruz Island (Fig. 1). Its climate corresponds to hot arid desert and tropical savannah (Beck et al. 2018), and its soils are mostly red, constituted by sand-clay-loam (Franz 1980; Poggio et al. 2021). The predominant vegetation on Baltra Island is classified as "nanophyllous desert scrub," primarily characterized by the prevalence of *Opuntia echios* (Cactaceae) cacti and shrubs such as *Castela galapageia* (Simaroubaceae). Additionally, it is described as having a sparsely populated herbaceous layer (Hamann 1975; INGALA et al. 1989). A past survey conducted by Alban Stewart in 1905 challenges the previous description. The author observed that the past vegetation on Baltra Island appeared to be denser, with evidence of regenerating trees, shrubs, and cacti dispersed throughout the landscape (Stewart 1911; Hamann 1979).

Baltra is the most visited island within the Galapagos archipelago, receiving more than 72% of tourist transit (Zander et al. 2016). Tourists are captivated by its astonishing dry landscapes (Fig. 1), but Baltra hides a history of vast degradation for the less informed visitors. The island's vegetation was negatively affected by the influence of introduced feral goats (*Capra hircus*) and cats (Balseca 2002; Phillips et al. 2005) for over a century (Slevin 1931; Phillips et al. 2012). Although these foreign animals have been eradicated on the island (since 2000 and 2005, respectively), plant populations have not yet recovered (Phillips et al. 2005; Negoita et al. 2021). Additionally, in World War II, a military base was constructed (Fig. S2), increasing perturbation through land use change (Woram 1991), and probably mediating the local extinction of land iguanas, the main seed disperser, during the 1940–1990s (Phillips et al. 2005). Land iguanas were repatriated to Baltra through several campaigns started in 1991 (Cayot & Menoscal 1994; Buitrón 2000), and the last census has estimated its population

in 781 individuals (Márquez et al. 2010). Nowadays, despite the reduction in military occupation extent, the considerable number of tourists passing through the island has escalated. The increase in tourism has led to a higher production of garbage, posing challenges for waste management on the island and resulting in the disposal of non-recyclable materials in waste-fillings (Muñoz-Pérez et al. 2023).

North Seymour Island

North Seymour (0°23'38"S; 90°17'06"W; Fig. 1) is an uninhabited island of less than 40 m.a.s.l. elevation, 1.46 km north of Baltra, covering an area of just 184 ha (Snell et al. 1996). Similar to Baltra, it has a hot arid desert climate (Beck et al. 2018), and soils that are mostly red, constituted by clay-loam (Franz 1980; Poggio et al. 2021). According to the literature, the North Seymour plant community is mainly composed of trees such as *Bursera graveolens* (Burseraceae), *O. echios* cacti, small shrubs such as *Croton scouleri* (Euphorbiaceae), and several forbs and grasses (Izurieta 1991).

North Seymour has the potential to be used as a semi-pristine reference ecosystem for Baltra because it has never been inhabited and was not invaded by goats or cats. Also, due to the proximity, it could be the closest to what Baltra would have been had degradation not occurred. Nonetheless, during World War II it was also used as a weapon testing area (Reynolds 1981), and before the 1944–1954 local extinction of land iguanas in Baltra, some individuals were transferred to North Seymour, which now homes the largest population of this locally non-native vertebrate (Tzika et al. 2008). This historical context serves as an indicator of North Seymour's potential as a suitable reference candidate. The last of these iguanas, almost lost due to the landscape destruction on Baltra, discovered an ideal environment on North Seymour, resembling their native habitat (Cayot & Menoscal 1994). All in all, North Seymour can be used as a low degradation reference ecosystem, as few other areas are available in the archipelago.

First Survey

In the 2021 warm-wet season, we set up 20 plots of 40 × 10 m in North Seymour to evaluate the species composition and density of woody plants (including cacti) (Fig. S3). As Baltra is a degraded landscape, the main aim of this survey was to address, which are the most dominant woody species in this reference ecosystem. Plots were established, avoiding the tourist area (18 ha), but randomly distributed across the remaining landscape. Each plot was subdivided into 16 subplots of 5 × 5 m, where we recorded each woody species and cacti, and their abundance (Calle-Loor et al. 2022). Abundance was registered, counting plants (i.e. stems) only if they were higher than 30 cm to avoid recording regenerating plants that could die posterior. Species were identified in the field or later in the local herbarium. To determine the influence of environmental factors on the occurrence of the plant species (as proposed by Durbecq et al. 2020), rock cover and slope were recorded visually inside each plot. To characterize the whole composition of

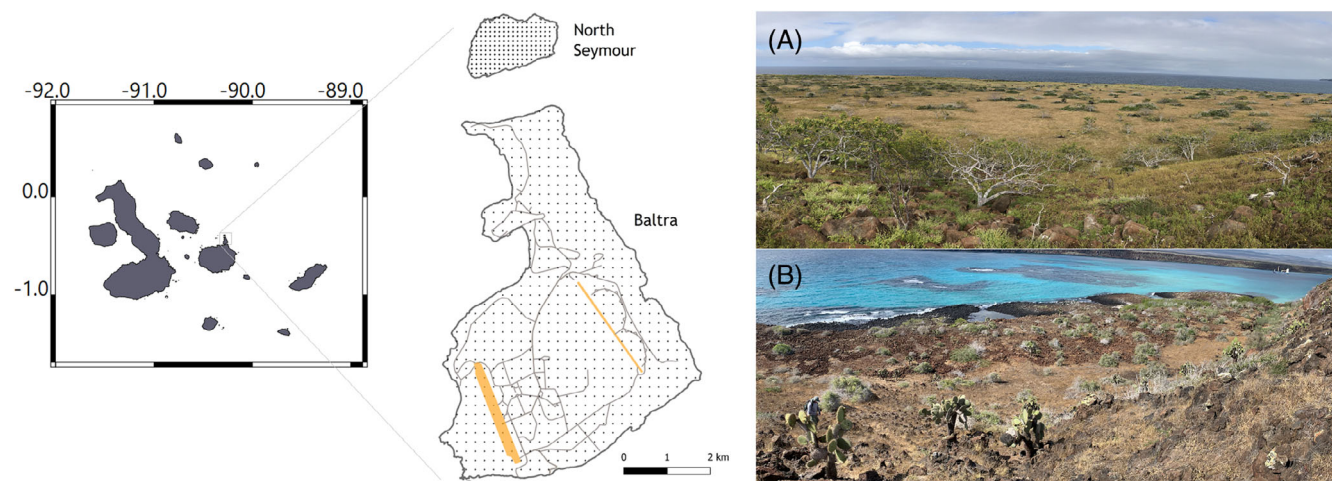


Figure 1. Location and landscapes of North Seymour and Baltra. Mosquera Island was deleted for simplicity. In the map, each dot corresponds to a plot on the main survey (2022). Orange rectangles in Baltra correspond to new and old flight tracks, which were not sampled. Photos: North Seymour (A), Baltra (B).

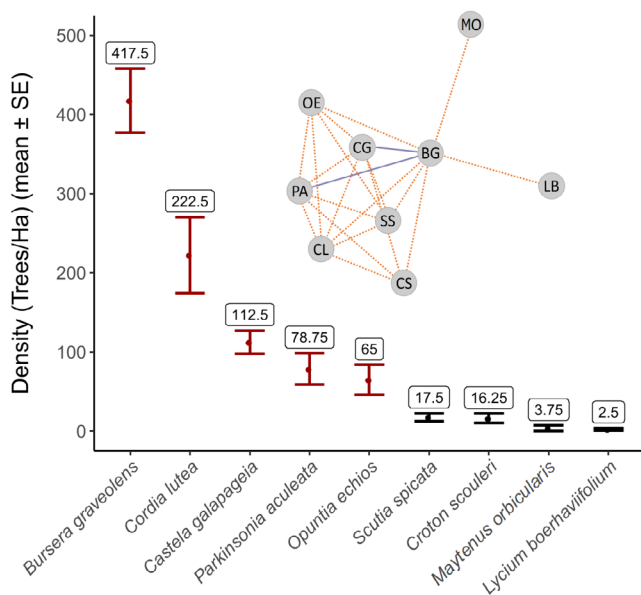


Figure 2. North Seymour woody species density and co-occurrence. In red, the five most dense species (dominant). Upper network shows all co-occurrence found in field. Dashed-orange lines: co-occurrence not different than random; blue lines: co-occurrences with a rate higher than expected by random ($p \leq 0.05$). All the abbreviations in the networks correspond to names in the plot.

North Seymour species, in addition to the plot survey focused on woody plants, all the other vascular plants were registered as a species list. The species screening was complemented by also searching for species in the periphery of the plots and in the tourist area.

Second Survey

After defining the five most dominant species in the reference ecosystem, we conducted a targeted second large-scale survey

in the 2022 cold-dry season, to establish the density and distribution of these species across the whole surface of both islands. To achieve this, we divided the islands into evenly spaced coordinates, creating a uniform grid for the survey. In North Seymour, we evaluated 178 plots, while in Baltra 607 (Fig. 1). Due to differences in area size and restrictions in resources (i.e. time and budget), in North Seymour, plots were located at every 100 m, while in Baltra at every 200 m. We specifically recorded the presence of the previously identified five most dominant (i.e. from the first survey analyses) within a circular area with a 20 m radius, using each coordinate as the center (Galapagos Conservancy & Dirección del Parque Nacional Galápagos 2022). Woody species were separated into two height classes: adults and juveniles (<50 cm tall), while *Opuntia* cacti were separated into three classes: adults (tall and developed trunk with bark and without spines), subadults (as the former but with spines), and juveniles (without trunk).

Analyses

First Survey. To evaluate which were the most dominant woody species in North Seymour, abundance data of the first survey was summarized per species to obtain density, using the area of the plot, and expressed as means with standard errors. Data at the subplot scale were converted into a presence-absence matrix, and the co-occurrence of species was analyzed, testing which pair of species occurred at a rate higher than expected by chance ($p \leq 0.05$). The occurrence of species was also evaluated through generalized additive models (GAM) using the environmental data (i.e. rocks cover and slope) as explanatory variables, with p -values ≤ 0.05 indicating significant associations. Through this, we addressed the structure of the dominant plant community and which environmental factors are preferred by species. Finalizing the first survey data, the whole North Seymour species list was compared to early

collects on the island to evaluate potential compositional differences.

Second Survey. As a proxy of the community resilience (i.e. the capacity to overcome degradation), maps of richness, and plants density across different height classes per species were created. For this, the data from the second survey were projected in QGIS 3.24 and rasterized. To obtain the densities for each height class, the abundances recorded on the circular area of 20 m radius were extrapolated to 1 ha. Final maps were generated with the same resolution as the survey (i.e. 100 × 100 m in North Seymour and 200 × 200 m in Baltra). Presence-absence data at the species level were used to estimate the frequency of plots containing each dominant species per island. Finally, and similar to the previous survey analyses, co-occurrence higher than expected by random ($p \leq 0.05$) was estimated in order to check if community structure in Baltra is kept compared to North Seymour.

In Baltra, we identified two specific regions characterized by a higher density of the single most dominant species, which may inform more pristine or resilient reference-areas within the island. We examined the co-occurrence network within these areas to evaluate if they were structured differently than the whole island. For this, in QGIS 3.24, the two areas were delimited creating a minimum convex polyhedron perimeter using the presence of this specific single species, and presence data for the five dominant species were extracted for each area. Through this procedure, we avoid using large areas of highly degraded landscape of Baltra, and rather evaluate if the co-occurrences of its slightly better conserved areas are more similar to those in North Seymour. Analyses were done in R Version 4.2.1 (R Core Team, 2023), with the packages “stats” for densities, “mgcv” (Wood & Wood 2015) for the GAM, and “cooccur” (Griffith et al. 2016) and “visNetwork” (Almende et al. 2022) for co-occurrences.

Results

First Survey

The first survey showed that nine woody species are frequently present in North Seymour (Fig. 2). The most dominant species included trees such as *Bursera graveolens*, *Cordia lutea* (Boraginaceae), and *Parkinsonia aculeata* (Fabaceae), shrubs such as *Castela galapageia*, and the cacti *Opuntia echios*. *Bursera graveolens*, being the dominant tree species, is the only species having co-occurrence with all other woody species (Fig. 2), and from these, the co-occurrence with *C. galapageia* and *P. aculeata* is higher than the expected by chance ($p = 0.018$ and $p = 0.003$, respectively).

Most GAM models for rock cover had a bad fit with the data, except for *B. graveolens* (deviance = 0.62; Table S2). However, when these models were assessed, essentially none were significant (all p -value > 0.05) (Fig. S4). The same was true when we analyzed the effect of slope.

The comparison between historical and current species lists of North Seymour showed that similar species were found already a century ago (Tables 1 & S1), with *Maytenus orbicularis* as the

only woody species not recorded by Stewart. Overall, both species lists had a high proportion of native and endemic species. The current species list detected three endemic and native species as highly frequent (≥ 0.7 ; but not abundant): *Althernanthera snodgrassii*, *Aristida subspicata*, and *Bouteloua disticha*. Stewart (1911) recorded 19 species, of which only 1 had doubtful origin (5.3%), while 4 of 35 (11.4%) were cryptogenic in our survey. Our comparison suggests that certain species may have been wrongly identified in the past but were matched to species from the new survey list at the genus level (i.e. *Desmodium*, *Euphorbia*, and *Sesuvium*), while others found in the past were now absent (i.e. *Amaranthus*, *Atriplex*, *Bouteloua*, *Evolvulus*, *Gossypium*, and *Pectis*). The current species list shows that the introduced *Cleome viscosa* is an abundant new element, not recorded a century ago (Table 1).

Second Survey

The second survey results showed that while most North Seymour plots had a high richness of the dominant species (i.e. 3–5 dominant species/ha), in Baltra this was true only for the west-south and north part of the island (Fig. 3). Instead, a diagonal from the west-center to the south-east was found to have lower richness (1–2 species/ha), with mostly only presence and high density of *C. galapageia* (Figs. 3 & S5). From the five most dominant species, only *P. aculeata* was slightly more frequent on Baltra (Table 2), while *C. lutea* was highly frequent on North Seymour (88%) but few plots contained its presence on Baltra (13%). The distribution of all species was mainly homogeneous throughout North Seymour, with the central area harboring the highest densities (Fig. S5). On the contrary, most species in Baltra had a disrupted distribution. One particularly important species in need of restoration due to its significance within the ecosystem is *O. echios* (Gibbs 2013). The analysis showed that while adults and subadults of this species are evenly distributed across the island, the presence of juveniles was sparse and scattered, with most areas showing low density (Fig. 4). This low regeneration pattern was found with most of the dominant species on both islands and was particularly clear in North Seymour (Fig. S5).

In the co-occurrence analyses, the only one that was significant across all networks was *B. graveolens* and *P. aculeata* (Table S3; Figs. 2 & 5) ranging from 0.2 to 0.43 probability of co-occurrence, depending on the island and area under evaluation (Fig. 5). The co-occurrence of dominant species in the second survey on North Seymour showed that *B. graveolens* and *C. lutea* are species co-occurring significantly higher than all the other species (Fig. 5). Yet, in Baltra, this structure is lost and no species is co-occurring significantly higher with more than two species (Table S3; Fig. 5). Finally, the split analysis of Baltra showed that the network is lost even if the focus is in the specific two smaller areas with a higher presence of *B. graveolens*.

Discussion

When working with archipelago ecosystems, it is often challenging to define reference models for restoration, particularly

Table 1. List of species found in the first survey in North Seymour. For each species the origin and habit are given. Relative frequency column corresponds to the ratio of plots with presence of the species. Frequency equal 0 means it were only recorded outside the plots.

Species	Origin	Habit	Relative frequency
<i>Alternanthera snodgrassii</i> (Robinson) Howell	Endemic	Shrub	0.7
<i>Aristida subspicata</i> Trin. & Rupr.	Endemic	Grass	1
<i>Bouteloua disticha</i> (Kunth) Benth.	Native	Grass	1
<i>Bursera graveolens</i> (B.L.Rob.) A.Weeks & Tye	Native	Tree	0.8
<i>Castela galapageia</i> Hook.f.	Endemic	Shrub	0.85
<i>Cenchrus platyacanthus</i> Andersson	Endemic	Grass	0.2
<i>Chloris virgata</i> Sw.	Cryptogenic	Grass	0
<i>Cleome viscosa</i> L.	Introduced	Forb	0.6
<i>Cordia lutea</i> Lam.	Native	Tree	0.8
<i>Croton scouleri</i> Hook.f.	Endemic	Shrub	0.35
<i>Cryptocarpus pyriformis</i> Kunth	Native	Forb	0
<i>Desmodium glabrum</i> (Mill.) DC.	Cryptogenic	Forb	0.1
<i>Distimake aegyptus</i> (L.) A.R.Simões & Staples	Native	Climbing vine	0.1
<i>Eragrostis cilianensis</i> (All.) Vignolo ex Janch.	Cryptogenic	Grass	0.2
<i>Euphorbia punctulata</i> Andersson	Endemic	Forb	0.05
<i>Exodeconus miersii</i> (Hook. f.) D'Arcy	Endemic	Forb	0.1
<i>Galactia striata</i> (Jacq.) Urb.	Native	Forb	0.05
<i>Heliotropium angiospermum</i> Murray	Native	Forb	0.05
<i>Heliotropium curassavicum</i> L.	Native	Forb	0
<i>Lycium boerhaviifolium</i> L.f.	Native	Shrub	0.25
<i>Lycium minimum</i> C.L. Hitchc.	Endemic	Shrub	0.05
<i>Maytenus orbicularis</i> (Humb. & Bonpl. ex Schult.) Loes.	Native	Tree	0.05
<i>Mollugo flavescens</i> Andersson	Endemic	Forb	0.05
<i>Neptunia plena</i> (L.) Benth.	Native	Forb	0.2
<i>Nolana galapagensis</i> (Christoph.) Johnst.	Endemic	Forb	0
<i>Opuntia echios</i> (Howell) E.F. Anderson & Walk.	Endemic	Tree-like cacti	0.45
<i>Panicum hirticaule</i> J.Presl	Native	Grass	0.25
<i>Parkinsonia aculeata</i> L.	Native	Tree	0.7
<i>Physalis angulata</i> L.	Native	Forb	0.1
<i>Portulaca howellii</i> (D. Legrand) Eliasson	Endemic	Forb	0
<i>Scutia spicata</i> (Hook. f.) M.C. Johnst.	Native	Shrub	0.55
<i>Sesuvium edmondstonii</i> Hook.f.	Native	Forb	0
<i>Trianthema portulacastrum</i> L.	Native	Forb	0.3
<i>Tribulus terrestris</i> L.	Cryptogenic	Forb	0.05
<i>Urochloa fusca</i> (Sw.) B.F.Hansen & Wunderlin	Native	Grass	0.1

in the absence of historical records and the dissimilarity to ecosystems outside these areas. In this paper, we analyzed the species composition and community structure of North Seymour vegetation, and through a second survey, we compared the dominant species with Baltra Island. The information compiled can serve as initial vegetation recovery targets and as a proxy for one of the reference ecosystems needed to define a larger reference model to restore Baltra island in the Galapagos Archipelago.

North Seymour might be one of the most appropriate reference ecosystems to restore Baltra due to its vicinity and low degradation. In fact, scientific literature suggests that both islands were connected once by land with the north-east coast of Santa Cruz, during Pleistocene glacial maxima (Franz 1980; Geist 1996) constituting a *stepping stone* lava plateau, together with Mosquera Island. Evidence supporting this is the lack of genetic distance between *Bursera graveolens* populations from North Seymour and Baltra (Weeks & Tye 2009).

Historical data on both islands are scarce (Stewart 1911; Alpert 1946), but the 2021 survey of our study shows that

several of the few species recorded earlier in North Seymour are still present. Also, the number of species present in Baltra was double that of North Seymour, as expected through island biogeographic theory (MacArthur & Wilson 2001). Due to their close proximity, they should hold similar species, but their richness must decrease with island size. This is also reflected in the number of plant species registered in the CDF collections for each island (94 vs. 72, respectively; CDF 2023). An interesting finding of the species list is that all of the dominant species defined by the current North Seymour community, were already recorded at the beginning of the twentieth century on Baltra. Albeit the previous densities in Baltra are unknown, this might indicate that North Seymour is a proper reference ecosystem. This background helps to depict a historical frame of the flora elements of Baltra, which now include exotic species. In fact, the appearance of *Cleome viscosa* in North Seymour is in tune with the presence of this introduced species in several islands of the archipelago (Guerrero et al. 2008; Traveset et al. 2013), Baltra included (Herrera & Causton 2010). Despite its widespread

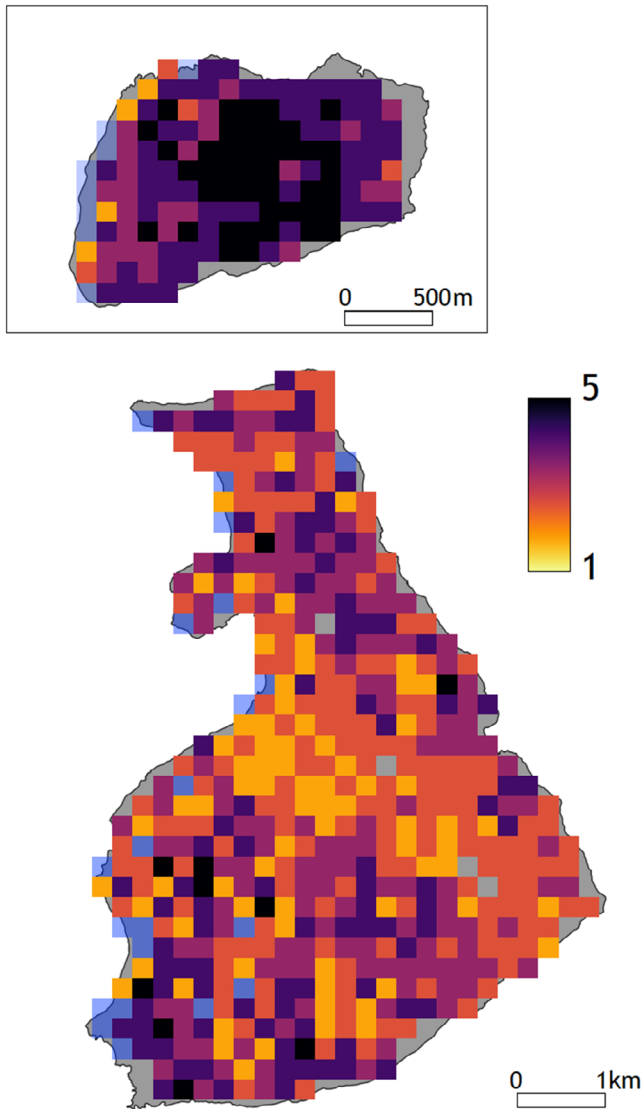


Figure 3. Rasterized richness of dominant species in North Seymour and Baltra islands. Richness values equal to 0 are presented in blue. Rasterized data have a pixel size of 200×200 m in Baltra and 100×100 m in North Seymour. North Seymour Island is at a $2\times$ scale for comparison means.

presence in Baltra and introduction to numerous islands globally, the literature only documents its negative impact on crops (Emami Kongor et al. 2019; Akbari-Gelvardi et al. 2021), with no studies investigating its effects on natural ecosystems. Then, without new studies, little can be recommended about its management in the Galapagos context.

Our first survey showed that the five most dominant species can reach mean densities from 65 to over 400 individuals per hectare in North Seymour, depending on the species. Values within this range have been previously proposed as historical densities for Baltra using similar undisturbed areas of the archipelago (Negoita et al. 2021). The 2022' survey reached higher density values, but both surveys found the same order of species with highest densities indicating that both approaches yielded

Table 2. Frequency of the five most dominant species per island. In each island the values represent the percentage of plots containing the species (178 plots in North Seymour and 609 in Baltra).

Species	Habit	Baltra	North Seymour
<i>Bursera graveolens</i>	Tree	37.6	89.89
<i>Castela galapageia</i>	Shrub	76.68	85.96
<i>Cordia lutea</i>	Tree	13.14	88.2
<i>Opuntia echios</i>	Tree-like cacti	59.11	69.66
<i>Parkinsonia aculeata</i>	Tree	53.85	47.75

similar trends. Albeit replicating the densities is not the final goal of the restoration, this information is crucial to guide planting efforts. For example, three out of the five dominant species were endemic (i.e. *B. graveolens*, *Opuntia echios*, and *Castela galapageia*), indicating the importance of considering them.

Species present at lower densities, such as *Scutia spicata*, *Croton scouleri*, *Maytenus orbicularis*, and *Lycium boerhaviifolium*, also need be considered on the restoration plan, as they might have important ecological roles. For example, species like the former have been mentioned to increase bird biodiversity at the archipelago scale (Traveset et al. 2015). In fact, a preliminary study has already assessed how to restore some similar taxa, such as *Lycium minimum* on Baltra island (Negoita et al. 2021). Interestingly, these species were also found outside of our North Seymour 2021' plots, but in lower density (<1 ind/ha). Consequently, by expanding the scope of restoration efforts to encompass these additional species, we could enhance the overall restoration goal for Baltra.

Izurieta (1991) defined two areas on North Seymour while studying land iguanas, based on differences in altitude (approximately 25 m): the "eastern platform" (EP) at the east of the island and the "high and rocky area" (HRA) to the center-west. The study found *O. echios* as the species with the second-highest coverage in EP, while the same was true for *Parkinsonia aculeata* for in HRA. Our second survey of North Seymour, indeed found *O. echios* as more represented to the east of the island, while *P. aculeata* more represent to the center-west, showing that these species probably find better conditions to inhabit those areas. A potential cause could be the soils or the wind direction, as we found a higher density of these species in similar cardinal directions on Baltra.

Our results show that regeneration is scarce in the studied ecosystem. Despite the Galapagos being mostly in a tropical climate (Beck et al. 2018), both studied islands overlap the arid savannah region. Usually, arid environments pose constraints to plant regeneration, and while plants can germinate in the rainy season, the scarce precipitation and lack of it for several months impede the posterior establishment (Chesson et al. 2004). This results in remarkably slow vegetation recovery trends in arid regions (Allen et al. 1995). Specifically, for the Galapagos Archipelago, the germination of seedlings is predominantly observed during specific years, mostly influenced by the occurrence of humid El Niño events (Tye & Aldáz 1999). Therefore, when planning restoration efforts in the region, it is crucial to

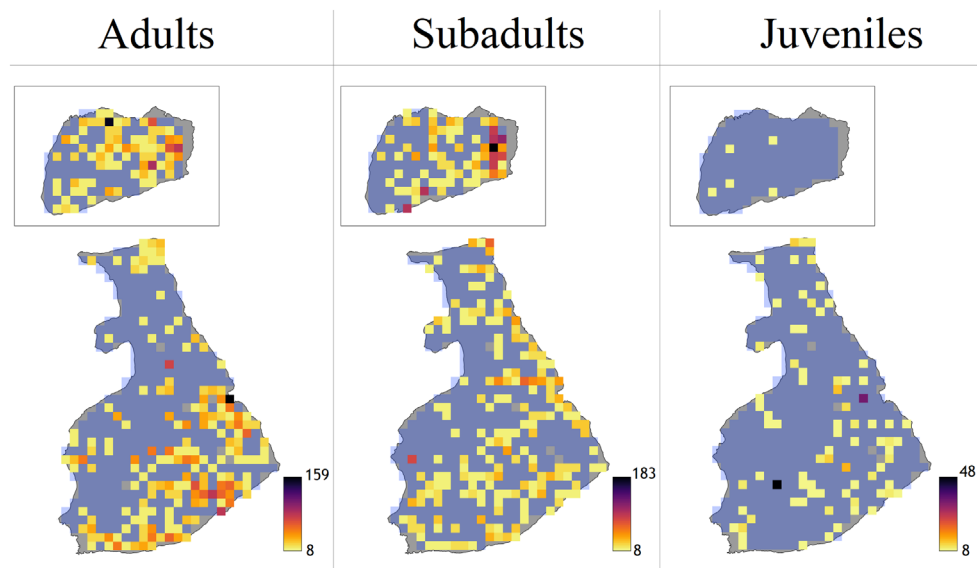


Figure 4. Distribution of height classes densities of *Opuntia echios* in North Seymour and Baltra. Different height classes maps have difference scales. Density values correspond to individuals per hectare. Values equal to 0 are presented in blue. Rasterized data have a pixel size of 200×200 m in Baltra and 100×100 m in North Seymour. North Seymour Island is at a $2 \times$ scale for comparison means. A, adult; S, subadults, J, juvenile. Density values according to circular sampled areas of 20 m radius.

adopt an opportunistic approach that considers these specific years (Gibbs 2013). Previous studies have shown that species such as *Opuntia* and *Bursera* have low regeneration (Hamann 2001). For example, even if goats have been eradicated in several islands, low regeneration of the whole *Opuntia* genus is still a pattern throughout the full archipelago (Grant & Grant 1989). Our results agree with this evidence, but also show that it is a general pattern as all the studied species have scarce and scattered juveniles. In fact, it has been proposed that the restoration of *Opuntia* cacti should consider implementing diverse spatial structures and propagule types (Hicks & Mauchamp 2000) to obtain better results. All in all, cacti and other species in the Galapagos require a significant investment in resources to secure their establishment, as water supply is the most common problem to tackle (Tapia et al. 2019; Negoita et al. 2021).

Reference models do not represent a state in the past and should consider historical and future change (Gann et al. 2019). In the case of Baltra, there are two important factors to be considered in the development of the model. Firstly, it should accommodate the existing infrastructure comprising an airport, roads, an Ecuadorian military base, and past and future renewable energy projects (Jiménez-Uzcátegui 2007; Energía Renovables para Galápagos 2010; Tecno-Ambiente 2021). The activities carried out in these areas will continue to have an impact on the ecosystem, potentially causing landscape fragmentation. The reference model can take this into account by increasing connectivity in alternative areas. Secondly, it should address the influence of climate change on species and community structure. An assessment of the vulnerabilities of the Galapagos to climate change found that the terrestrial ecosystems of the archipelago will experience more frequent and intense ENSO events

(Sachs & Ladd 2010; Banks et al. 2011). The increase in rain events will reduce the competitive advantage of arid-adapted indigenous species and make more favorable conditions for invasive species (Sachs & Ladd 2010; CI et al. 2011). Moreover, during El Niño events, *Opuntia* and *Scalesia* are known to face elevated mortality and recruitment rates (Tye & Aldáz 1999). Both genera, represented by *O. echios* and *S. crockerii*, are found on both Baltra and North Seymour. However, *Scalesia crockerii* may not have been detected during our first survey on North Seymour because it grows on cliffs, often in inaccessible areas. To maintain ecosystem functionality in the face of these changes, the reference model should consider potential shifts in species distributions and densities, by incorporating ecological analog species. Recent literature underscores the importance of considering ecosystem functions in restoration efforts (Kollmann et al. 2016).

Our study shows that the distribution of dominant taxa in Baltra is dependent on the species. Particularly, species such as *B. graveolens* and *P. aculeata* have disrupted distributions, potentially because of the effects of the military base (Woram 1991) and the current airport in the middle of the absence area. Interestingly, these species were the only two that had a significant co-occurrence in all analyses. Other studies have found other positive associations too, between *B. graveolens* and *O. echios*, with most juveniles of the first under the branch of the later (Clark & Clark 1981). This association was one of the few that was kept when North Seymour co-occurrence was compared to Baltra in the second survey. Other species, such as *Cordia lutea* have low frequency and density, suggesting that they are more prone to the alteration of the system. On the contrary, *C. galapageia* has a wide-ranging distribution with higher density in the most degraded area, suggesting a pioneer strategy, or

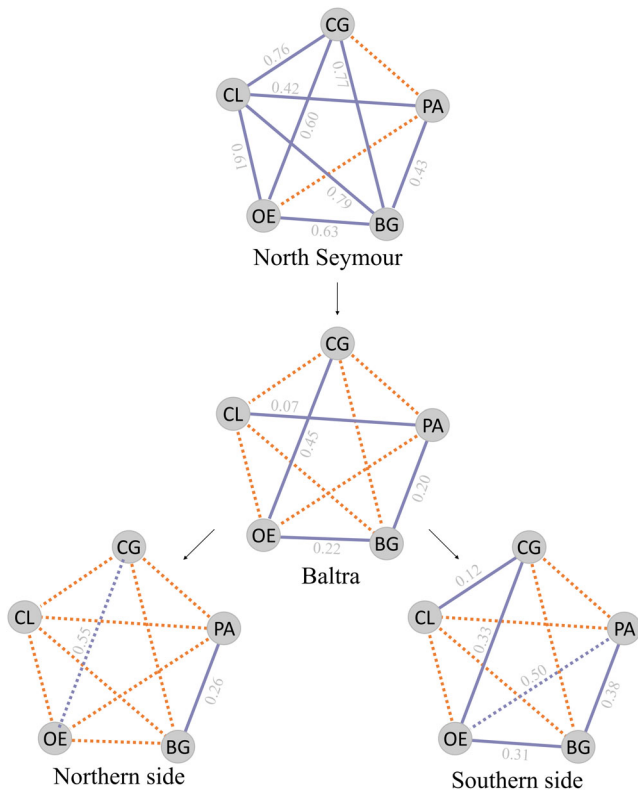


Figure 5. Co-occurrence structure of dominant species in North Seymour and Baltra. Dashed-orange lines: co-occurrence not different than random; blue lines: co-occurrences significantly higher than expected by random ($p \leq 0.05$). Dashed-blue lines; co-occurrences weakly significant ($\sim p = 0.06$). Gray numbers indicate the significant and weakly significant co-occurrence probability. BG = *Bursera graveolens*, CG = *Castela galapageia*, CL = *Cordia lutea*, OE = *Opuntia echios*, PA = *Parkinsonia aculeata*. Northern and Southern sides correspond to the two areas identified with high presence and density of *B. graveolens*.

as in other studies, that it was avoided by goats foraging (Hamann 1975). Our large-scale approach shows that the co-occurrence of dominant species assessed in Baltra is abruptly different from North Seymour, even if the focus is on areas where the most dominant *B. graveolens* is highly present, suggesting that the community structure of Baltra might have been lost. In light of this, the data suggest that Baltra needs actions to restore this co-occurrence structure. The GV2050 plans for restoring the island include the creation of a dispersal corridor to link the south-western and north-eastern areas that are disconnected (Gibbs 2013, 2016), which may help restore the co-occurrence network. New restoration actions planned for Baltra should implement the findings of this study, aiming to achieve the recovery targets of the arid plant communities and their structure. At the same time, the new restoration plan will expand the relevance of the study by using additional reference ecosystem within the archipelago to define a more concise reference model.

In summary, this study offers significant insights into Baltra Island's restoration challenges and opportunities, presenting a

transferable framework for vegetation analysis and reference ecosystem consideration.

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Supporting Information

The following information may be found in the online version of this article:

Figure S1. Overview of Baltra interactions.

Figure S2. Reference ecosystems (RE) and Short-Term Planning for Baltra's Reference Model (RM).

Figure S3. Areas sampled in the 2021' sampling.

Figure S4. GAM models for rock cover and species presence in North Seymour.

Figure S5. Distribution of densities by height classes for the dominant species.

Table S1. Oldest species list from Baltra and North Seymour, gathered from "Stewart (1911), *A botanical survey of the Galápagos Islands*, California Academy of Sciences."

Table S2. Summary of the GAM models for rock cover and slope.

Table S3. Significance of the co-occurrences between dominant species on North Seymour, Baltra islands and two areas of Baltra with high presence of *B. graveolens*.

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